

XMBA 200S: Data and Decisions

Chapter 23: Multiple Regression – Part 2

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Course overview
Part 4: Building regression models

- Blocks 3 and 4
- You want to build a model to forecast outcomes from data you have
- What's the best model you can build?
- This class: wrapping up multiple regression
- After break: building better models

Outline

- Wine Case
- A bit more about multiple regression

Exhibit B: Descriptive Statistics, Bordeaux Wines and Weather Variables

Variable	Mean	Standard Deviation	Minimum	Maximum
Price Per Bottle (in U.S. 1991 dollars)	$28.8	$21.0	$10.1	$100.0
Natural Log of Price	3.15	0.63	2.32	4.61
Age of Vintage (as of 1991)	24.2	8.2	11	39
Average Temperature over Growing Season from April to September (in degrees Celsius)	16.5	0.7	15.0	17.7
Rain During Harvest Season from August to September (in millimeters)	145	73	38	292
Rain in the Months Preceding the Growing Season from October to March (in millimeters)	608	129	376	830

Note: This table describes the vintages from 1952 to 1980, excluding the 1954 and 1956 vintages, which are now rarely sold. There are 27 total observations.

Exhibit C: Regression of the Price (in logs) on Weather Variables

Independent Variables	Estimate	Standard Error	*t*-statistic	p-value
Age of Vintage (as of 1991)	0.024	0.007	3.33	.003
Average Temperature over Growing Season from April to September (in degrees Celsius)	0.616	0.095	6.48	.000
Rain During Harvest Season from August to September (in millimeters)	-0.0039	0.0008	-4.78	.000
Rain in the Months Preceding the Growing Season from October to March (in millimeters)	0.0012	0.0005	2.42	.024
Intercept	-7.7	1.68	-4.61	.000

Note: All regressions use as data the vintages from 1952 to 1980, excluding the 1954 and 1956 vintages, which are now rarely sold. There are 27 total observations. The R^2 from the regression is 0.83 and the standard deviation of the residuals (s_e) is 0.29. The F-statistic for the model is 26.39 with a *p*-value of .0000.

Exhibit D: Bordeaux, France Weather Data for Years 1981 to 1988

	Age (as of 1991)	Avg. Temperature over Growing Season (Apr-Sep)	Rain During Harvest Season (Aug-Sep)	Rain Preceding Season (Oct-March)
1981	10	17.0	111	535
1982	9	17.4	162	712
1983	8	17.4	119	830
1984	7	16.5	119	591
1985	6	16.8	38	744
1986	5	16.3	171	563
1987	4	17.0	115	452
1988	3	17.1	59	808

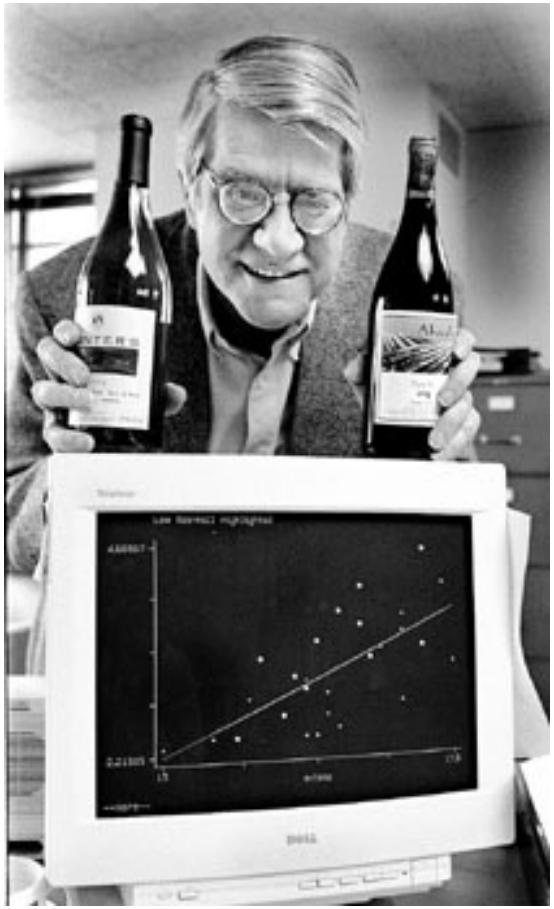
Note: All temperatures are in Celsius. Precipitation is measured in millimeters.

Exhibit E: Bordeaux Predictions, 1981-1988

	Prediction $\hat{y}$ (in logs)	95% Prediction Interval (in logs)	Prediction $\hat{y}$ (in dollars)	95% Prediction Interval	Picked By Experts
1981	3.22	2.64, 3.80	$25	$14, $45	No
1982	3.46	2.88, 4.04	$32	$18, $57	Yes
1983	3.76	3.18, 4.34	$43	$24, $77	No
1984	2.88	2.30, 3.46	$18	$10, $32	No
1985	3.54	2.96, 4.12	$34	$19, $62	Yes
1986	2.47	1.89, 3.05	$12	$7, $21	Yes
1987	2.96	2.38, 3.54	$19	$11, $35	No
1988	3.65	3.07, 4.23	$38	$21, $68	No

The prediction intervals were calculated using $\hat{y} \pm 2s_e$. See Stine and Foster, 2nd Edition, p. 553 for details.

Wine Case



The New York Times

Wine Equation Puts Some Noses Out of Joint

By PETER PASSELL
Published: March 04, 1990

Robert M. Parker Jr., generally regarded as the most influential wine critic in America, calls Professor Ashenfelter's research "ludicrous and absurd."

William Sokolin, a New York wine merchant, said the Bordeaux wine industry's view of the work ranges "somewhere between violent and hysterical." Those reactions are not surprising. No formula could capture the variations in estate-bottled wines from the same vintage. But if widely accepted, Professor Ashenfelter's approach to judging the quality of the red wines of Bordeaux and Burgundy long before they are drinkable could turn the clubby, exclusive world of wine rating on its ear.

Outline

- Wine Case
- A bit more about multiple regression

Residuals Review

Quick Review

Recall that residuals are the vertical deviations from the data points to the fitted line, $e = y - \hat{y}$.

The regression equation in a MRM is,

$$\hat{y} = b_0 + b_1 x_1 + b_2 x_2 + \ldots + b_k x_k$$

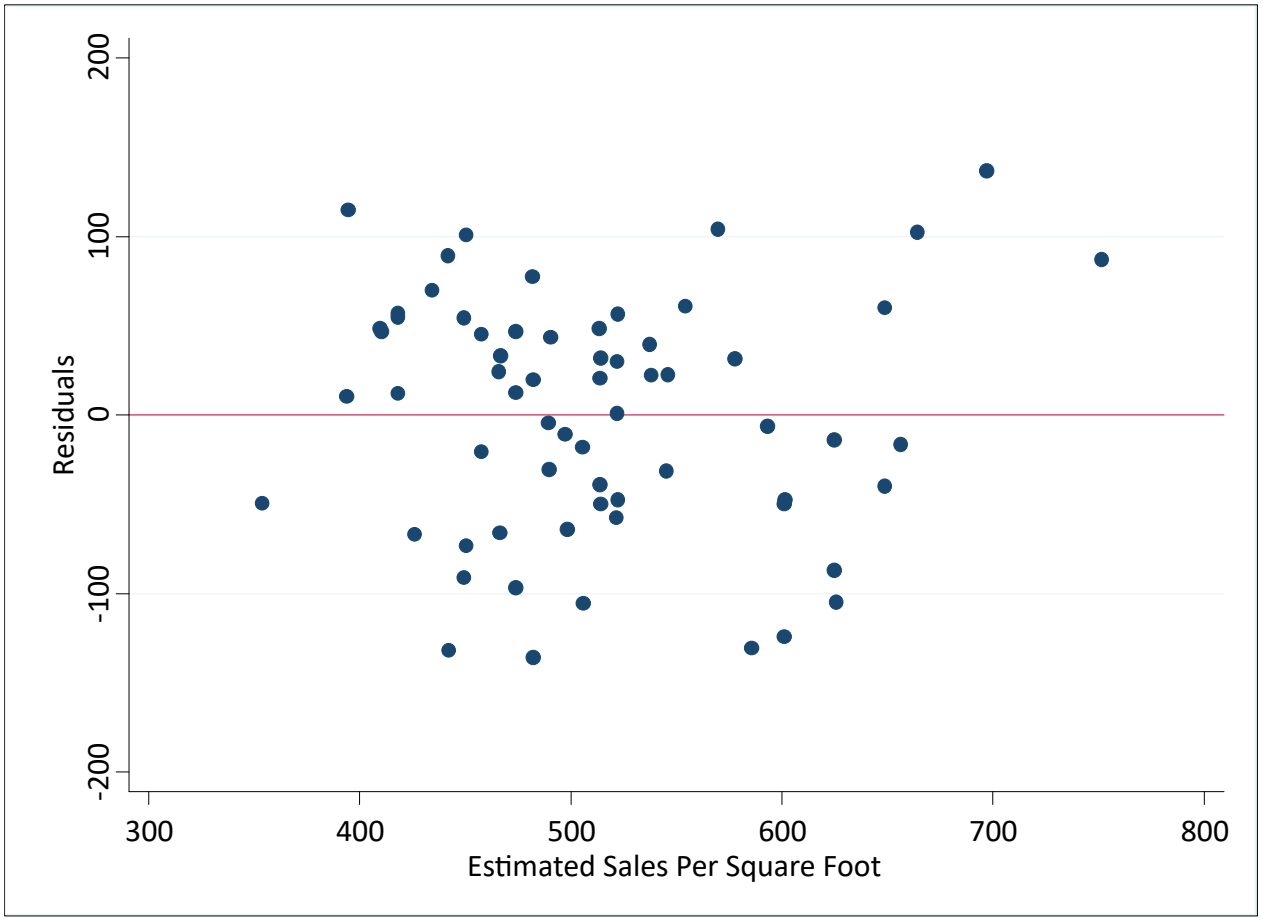
So, once we have estimated the intercept and slopes, we can calculate residuals.

- Note: EXCEL or Statistical Software does this for you

Peet's Coffee

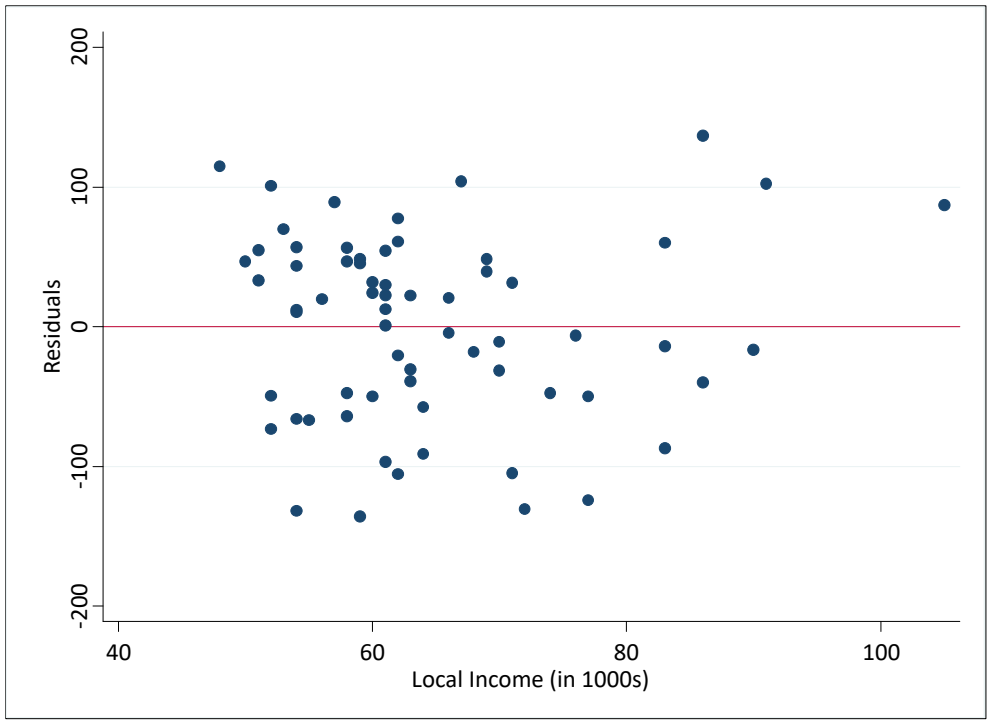


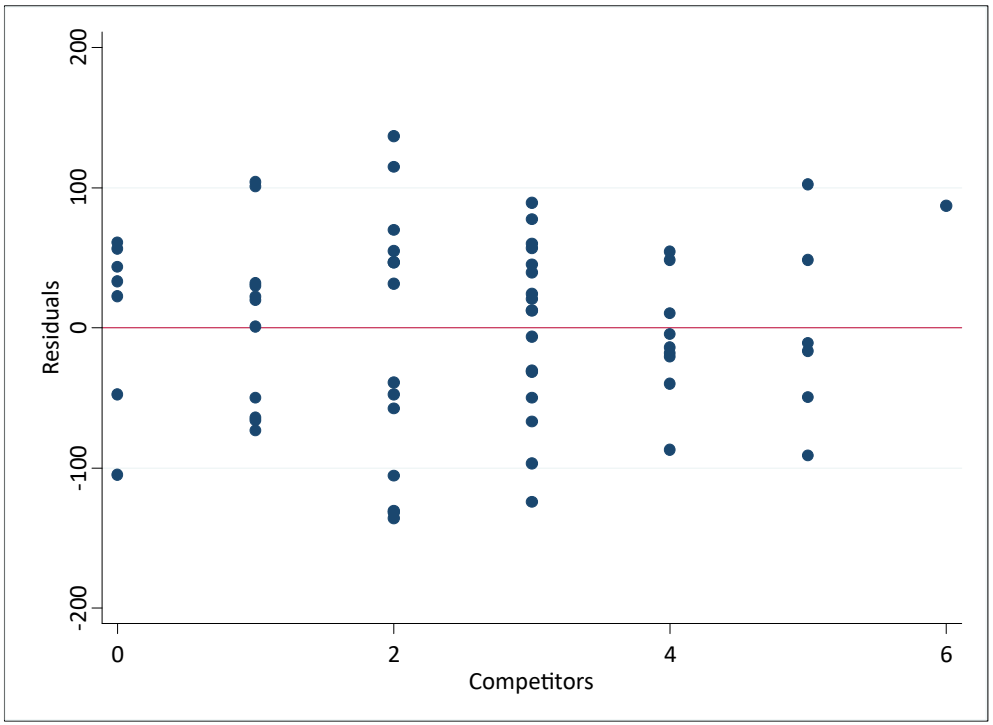
Residual Plot



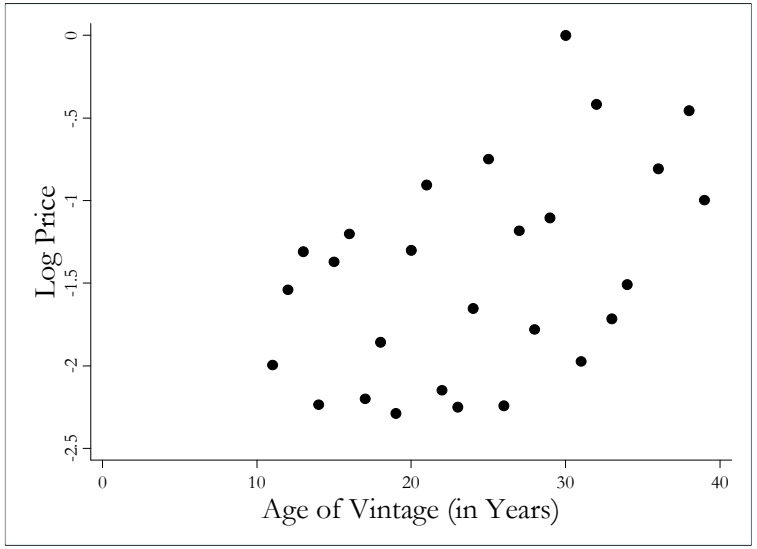
Peet's Coffee

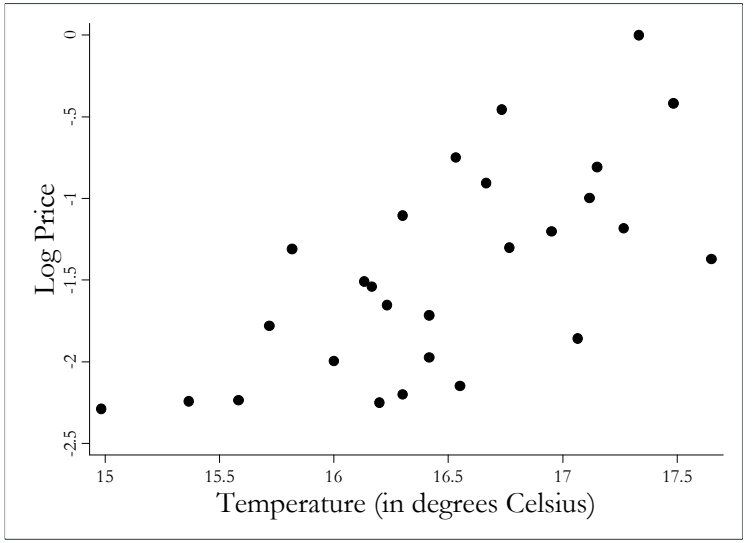
More Residual Plots

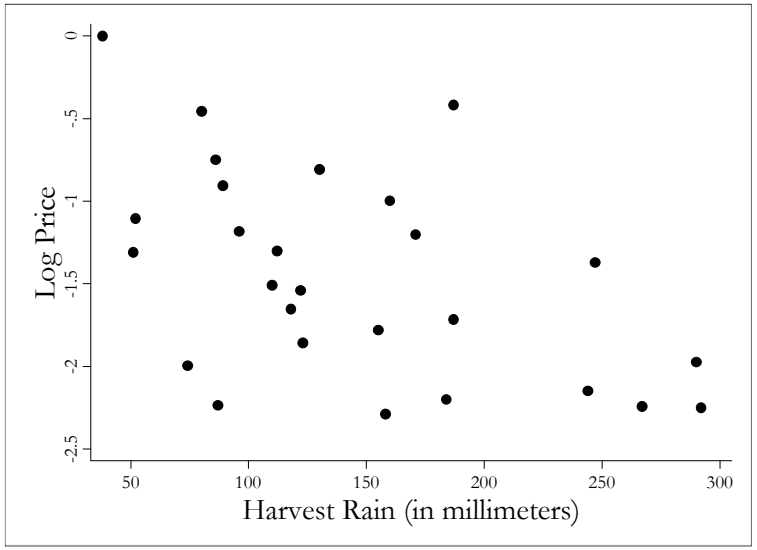


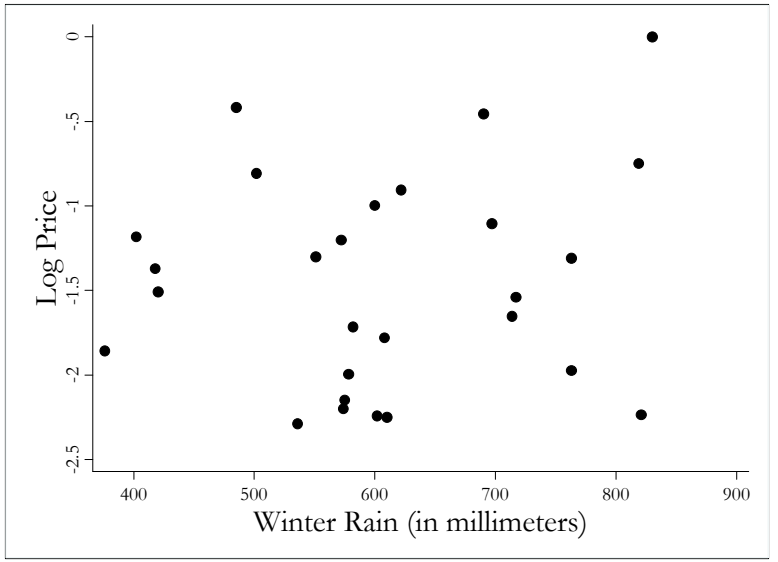


Wine Case, Scatterplots



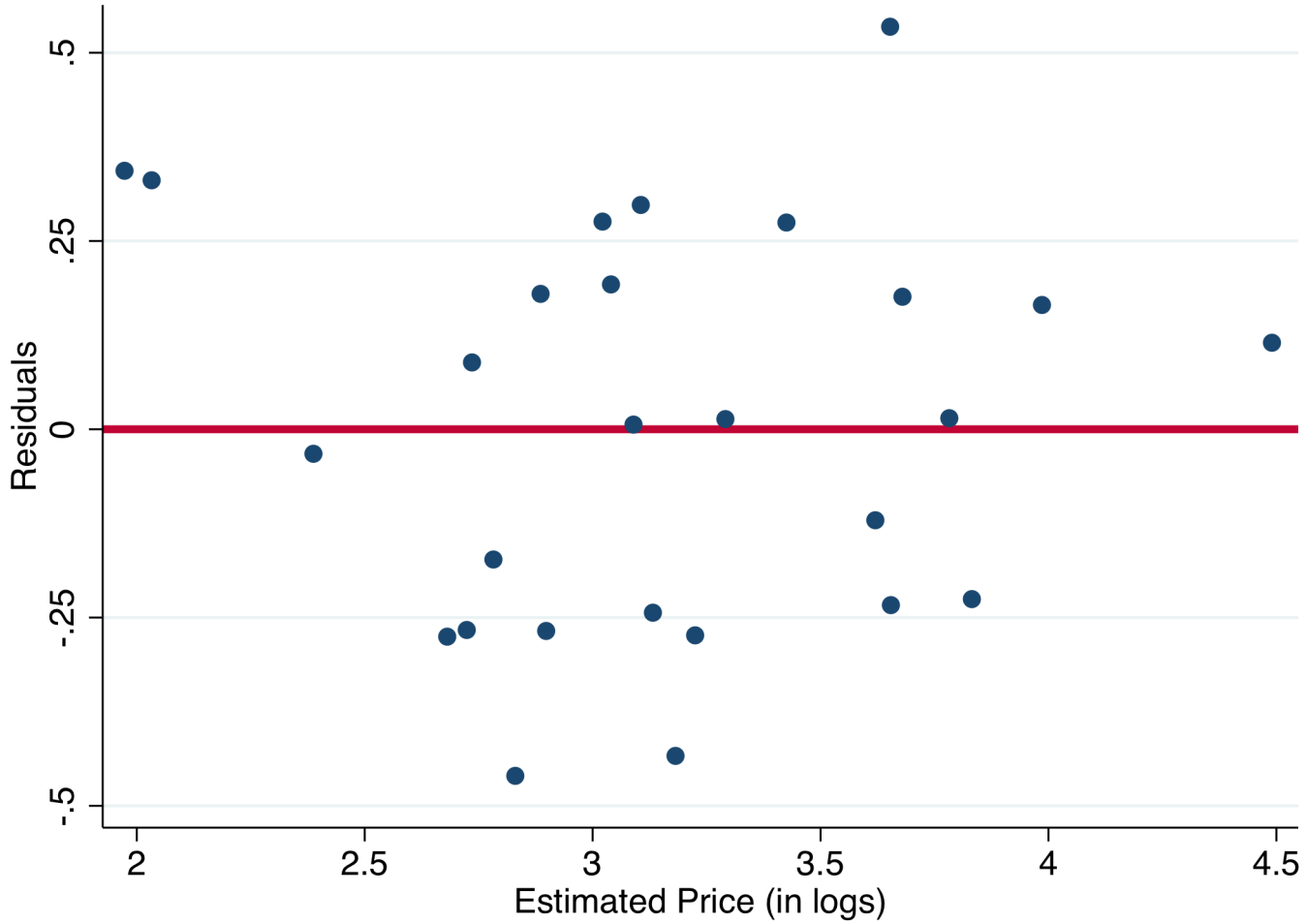






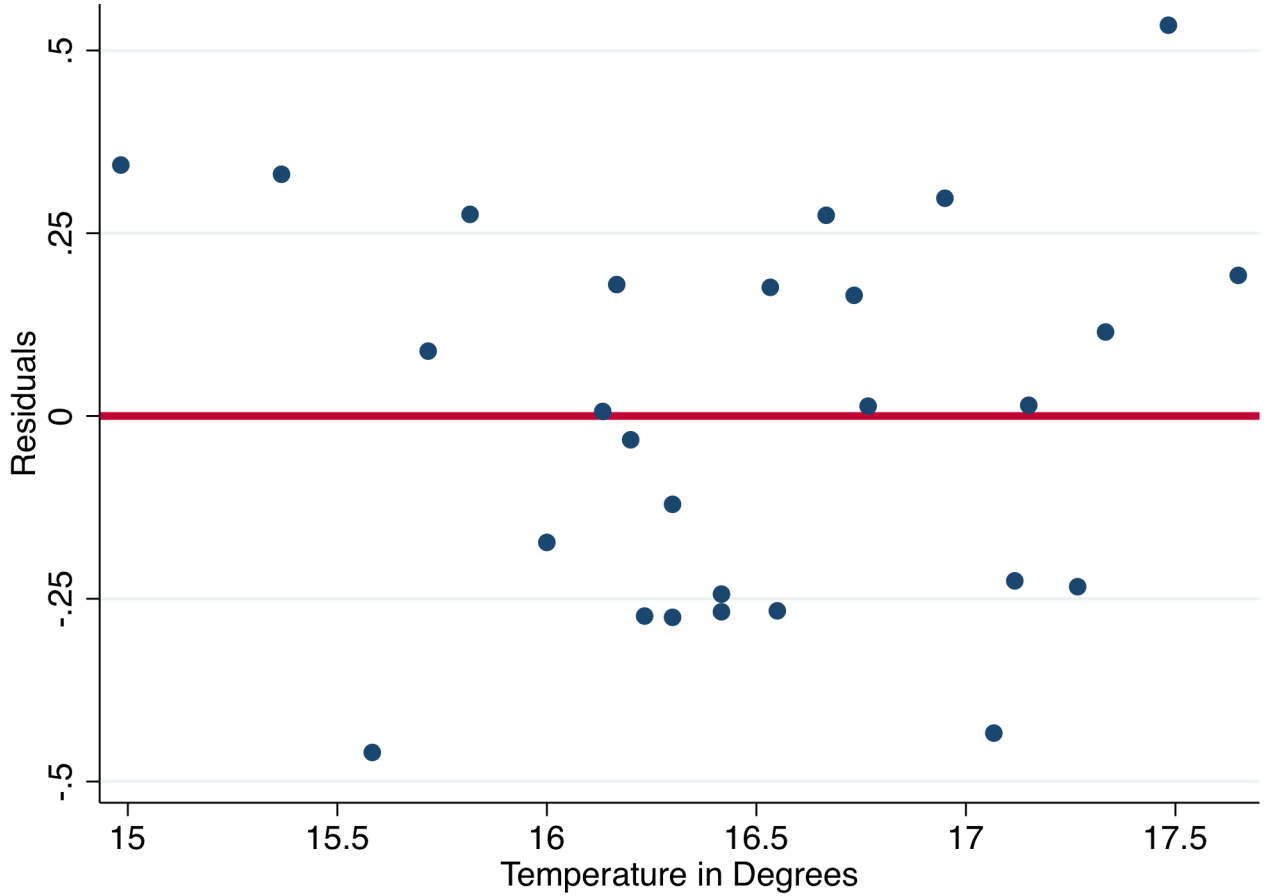
Wine Case

Residual Plot



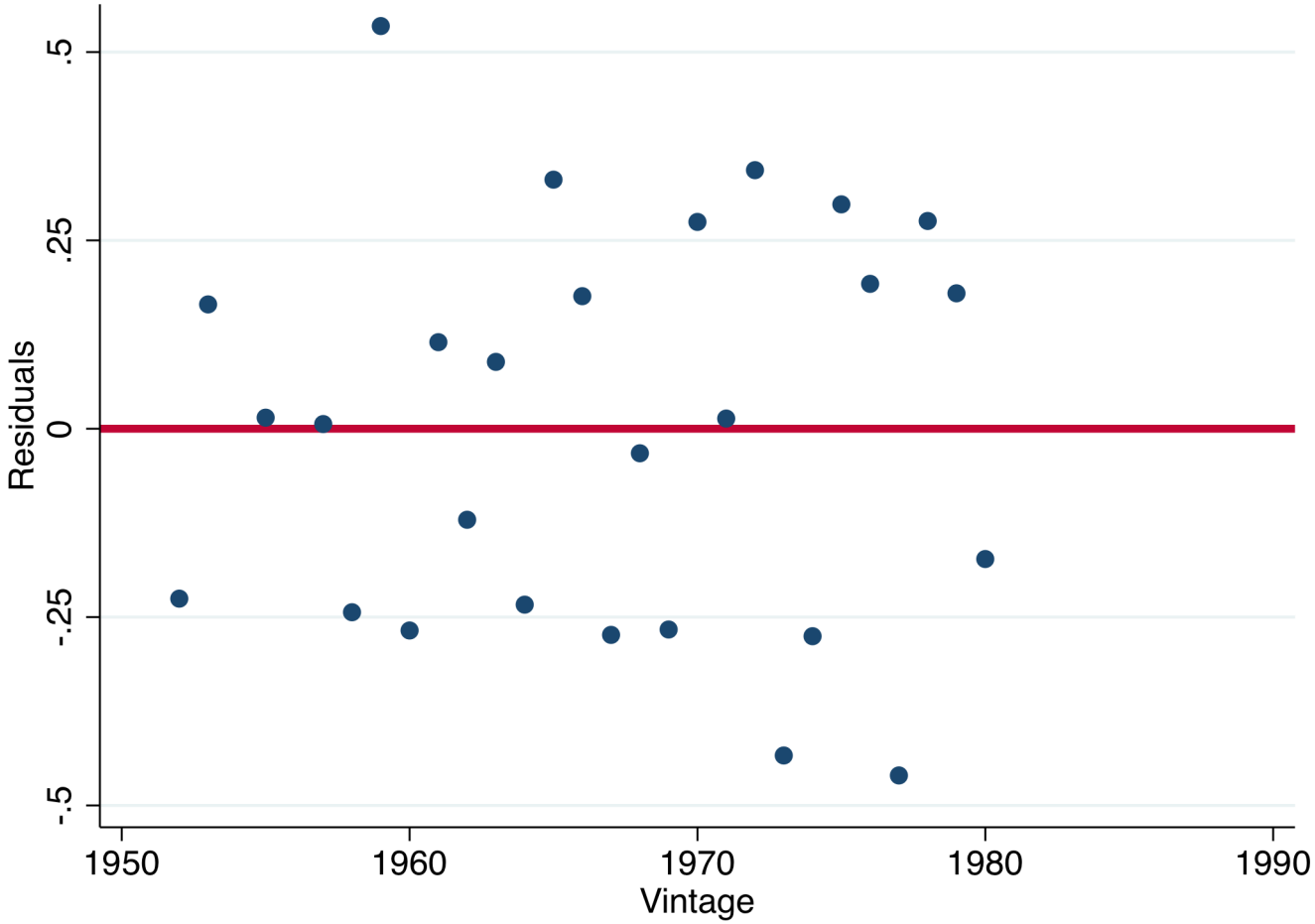
Wine Case

Residual Plot vs. Temperature



Wine Case

Residual Plot vs. Vintage



Inference in multiple regression

F-statistic

- The *F*-statistic is a measure of the overall explanatory power of a regression model

$$F = \frac{R^2}{1 - R^2} \frac{n - k - 1}{k}$$

- Used to test the null hypothesis that all slopes are equal to zero, e.g.,

$$H_0: \ \beta_1 = 0, \ \beta_2 = 0, \text{ etc.}$$

- Software automatically reports the *F*-statistic and associated *p*-value

- Interpretation: does you model explain anything related to the outcome variable (more than would be expected by random chance)

Inference: Peet's coffee

R^2	0.5947
$\overline{R}^2$	0.5816
s_e	68.0306
n	65

Source	Sum of Squares	df	Mean Square	*F*-Statistic	*p*-Value
Model	421,107.76	2	210,554	45.4940	<.0001
Error	286,946.30	62	4,628		
Total	708,054.06	64			

Term		Estimate	Std Error	*t*-Statistic	*p*-Value
Intercept	b_0	60.3587	49.2902	1.22	0.2254
Income	b_1	7.9660	0.8383	9.50	<.0001
Competitors	b_2	−24.1650	6.3899	−3.78	0.0004

- F-stat is large and corresponding p-value very low…
- The *t*-stats and associated *p*-values indicate that both slopes are significantly different from zero
- What else can you say?

F-Statistic

Peet's Coffee (SRM with Competitors)

SUMMARY OUTPUT

Regression Statistics	
R Square	.0044
Adjusted R Square	-.0114
Standard Error	105.78
F-Statistic	0.28
F-Statistic (*p*-value)	0.598
Observations	65

	Coefficients	*Standard Error*	*t-Stat*	*P-value*
Intercept	502.2	25.44	19.74	0.000
Competitors	4.635	8.75	0.53	0.598

P-value from this F-test is large. Fail to reject the null hypothesis that the variables in our model are unrelated to the outcome variable (sales)

Putting this into practice

How to “read” multiple regression output?

- Check the overall F-statistic before looking at the t-statistics: is there some explanatory power?
- What makes a variable important?
  - Statistical significance
  - Economic significance
- Keep in mind interpretation of the partial slopes estimated
  - Ceteris paribus – holding other factors fixed
- Always be careful in making statements about causality:
  - statistical significance (or high R-squared) doesn’t imply causality
  - controls don’t mean there are no other lurking variables

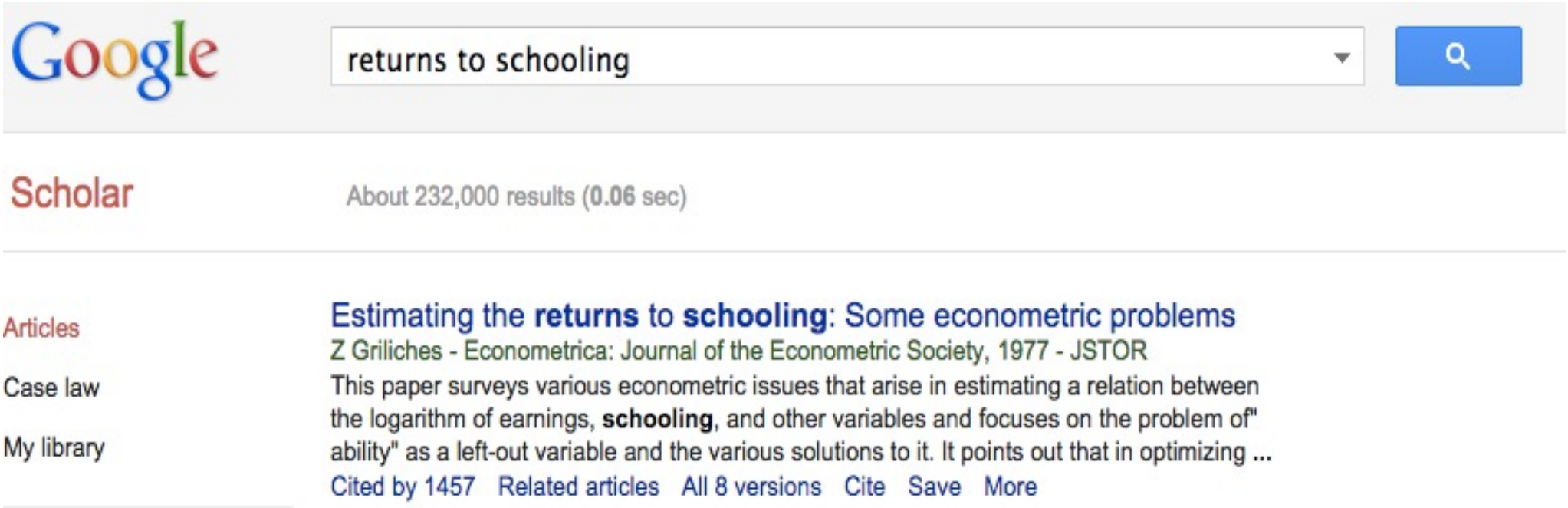
Takeaways

Use residual plots to look for nonlinearities and or other potential outliers driving model estimates

F-statistics can be useful overall model diagnostic before looking at any individual t-statistics

Bonus Material: Partial Slopes, and Bias from "Lurking Variables"

The “returns to schooling”



How much does an additional year of schooling earn?

The “returns to schooling”

- You have a dataset of individual incomes and demographic information
  - For each individual, you know yearly income, years of schooling, IQ, etc.
- Consider two different models:
  - SRM in which you regress income on years of schooling
  - MRM in which you regress income on years of schooling and IQ
- In which model will you get a larger estimated “return to schooling”?
  - Option A: In the SRM
  - Option B: In the MRM

Bias of Lurking Variables

Recall that MRM is a generalized version of the SRM. What happens in an SRM when we do not control for X_2?

$$Y = \beta_1 X_1 + \underbrace{\beta_2 X_2 + \cdots}_{\epsilon}$$

Recall what the equation of our statistic b_1 is:

$$b_1 = r\frac{s_y}{s_{x_1}} = \frac{cov(Y, X_1)}{s_{x_1}^2}$$

If we estimate the SRM when the true model is MRM then our estimate of b_1 will be biased. To see this, plug in the true "Y" into our b_1 formula

$$b_1 = \frac{cov(Y, X_1)}{s_{x_1}^2} = \frac{cov(\beta_1 X_1 + \beta_2 X_2 + \cdots, X_1)}{s_{x_1}^2} = \frac{\beta_1 cov(X_1, X_1) + \beta_2 cov(X_2, X_1) + \cdots}{s_{x_1}^2}$$

$$b_1 = \beta_1 + \underbrace{\frac{\beta_2 cov(X_2, X_1) + \cdots}{s_{x_1}^2}}$$

Bias from excluded/ missing lurking variables

Bias of Lurking Variables

- More generally, whenever there is any variable in the error term (i.e. a variable we do not control for) that is correlated with both:
  1. A variable we do control for, and
  2. The outcome variable

  Our estimates will be biased

- Bias may be ok if we are using regression to forecast/predict

- However, when we are interested in causal effects, bias is a problem
  - Often times we can say something about the direction of bias based on the likely correlation between the excluded and included variables

- Keep this in mind when interpreting reports, news articles, or regression estimates!

Bonus Material: Is this a “**Prediction**” Problem or a “**Causal Inference**” Problem?

What are Examples of Causal Questions?

Decision: Should I go to the hospital for treatment?
- Lots of people die in hospitals
- Are hospitals bad for health?

Decision: how much should I spend on advertising?
- Advertise more in December, sell more in December
- Is it because of advertising or other confounding festivities?

Decision: should we pay our employees more?
- More productive employees earn more
- Does that mean that higher pay *causes* higher productivity?

Decision: should I raise the price on a product?
- Positive relationship between price and sales
- Does that mean that higher prices *cause* more people to buy our product?

Causality: **Always** Think of the Counterfactual

Hospital:
- If I go to the hospital, will I be better off than I would have been if I didn't go?

Advertising:
- What would my sales be, if I had advertised less?

Pay Policy:
- What would productivity be, if I did not raise pay?

Sales:
- What would sales be, if I did not raise the price?

To develop a counterfactual, you need a control group
i.e. you need an experiment

What are Examples of Prediction or Forecast Problems?

Prediction: How productive will this person be if we hire them?
- Use data from similar employees to predict performance of this individual

Prediction: What is probability of dying from a heart attack based on most recent ECG?
- Use ECG data linked to death certificates to identify patterns and probabilities (see bCourses)

Prediction: What will our sales look like in December?
- Based on past sales and sales environment, use this data to predict/forecast December sales

Prediction: How many vacation days will my employees take this month?
- Based on previous vacation days in previous years, let's forecast vacation this month

What do you want to estimate?

Causal Impact:

- Change in sales associated with change in advertising expenditure, *holding everything else constant*

Prediction:

- Change in sales you would expect to observe when *advertising expenditure changes*

If you want to make a decision, the former is often the most relevant.
The latter only describes an association.

Example: iPads in the classroom



iPads in the Classroom

You can have a great model of the relationship between school investment in technology & test scores, but it won't answer question of what happens if you intervene and increase investment in technology. Why?

- Observed relationship likely not causal

- Observed data may show "more technology, higher test scores", but you want to know what happens to scores when you add iPads (i.e. their causal effect)

Estimating Demand: What is the intended use of demand estimation?

Tell consumers what to expect prices to be in the future?

- Want to model historical relationship
- Google flights to predict/forecast price likely to increase/decrease

Tell managers what to expect if they change price

- This is a counterfactual question
- Want to model causal relationship
- Ideally run an experiment
- If we increase prices, quantity sold will fall…

